

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

THE STATIONARY DISTRIBUTION OF A RANDOM WALK

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Recall: Sampling by Random Walk

Want to sample from a distribution P over some space $\mathcal{X}$.

- Start at any $x \in \mathcal{X}$
- Repeatedly move to a “nearby” state, with some transition probabilities
- After a while: the distribution over the current location is (close to) P

Random walk on a finite space $\mathcal{X}$

- Let Q_t be the position ("state") at time t
Next state Q_{t+1} depends only on the Q_t , not prior history: **Markov chain**
- Random walk is defined by $|\mathcal{X}| \times |\mathcal{X}|$ transition matrix

$$M(x, x') = M_{x, x'} = \Pr(Q_{t+1} = x' | Q_t = x)$$

- Let π_t be the distribution of Q_t , so $\pi_t \in \Delta_{\mathcal{X}}$

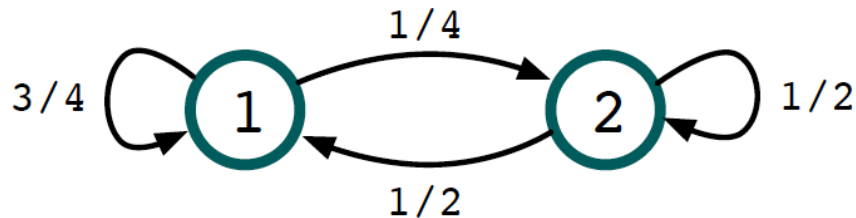
$$\pi_{t+1}(x) = \Pr(Q_{t+1} = x) = \sum_{x' \in \mathcal{X}} \Pr(Q_t = x') M_{x', x} = \sum_{x'} \pi_t(x') M_{x', x}$$

In vector form: $\pi_{t+1}^T = \pi_t^T M$.

We say π is a **stationary distribution** if $\pi^T = \pi^T M$.

Stationary Distribution

We say π is a stationary distribution if $\pi^T = \pi^T M$. Such a distribution always exists.
Determine the stationary distribution of this Markov chain:



When is there a Unique Stationary Distribution?

Theorem. Suppose there exists some $k > 0$ such that $M^k(x, x') > 0$ for all x, x' .
Then there is a unique stationary distribution π^* .

What we need:

- Under the random walk, it is possible to get from any location to any other location.
- Moreover, there is some number of steps k at which there is a non-zero probability of moving from any location x to any other location x' .

Figuring out the Stationary Distribution

Theorem. If π satisfies the **detailed balance** condition:

$$\pi(x)M(x, x') = \pi(x')M(x', x) \quad \forall x, x' \in \mathcal{X}$$

then π is a stationary distribution of M .